

Quantum Technologies Research Project

June Report

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Project Goals & Metrics

Research Gap: No unified comparison of BB84 vs MDI-QKD across key metrics

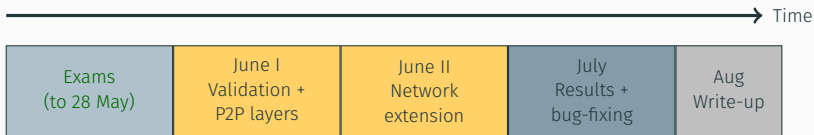
**Key Rate vs
Distance**
(varying networks)

**Implementation
Cost**
(economic & energetic)

Scalability
(user count)

- Analyse both protocols across varying network topologies
- Identify parameter regimes & topologies where each excels
- Inform practical QKD deployment decisions

Where We Were



- **June first half:** Validation benchmarking (BB84 + MDI); complete P2P modelling layers (dark counts → node loss → beyond)
- **June second half:** Begin network-scale extension; trusted node BB84 vs MDI comparison
- **July:** Bug-catching, parameter sweeps, result collection
- **August:** Write-up, final figures, submission preparation
- **September:** Viva

Since last meeting

May/June Progress

- P2P modelling layers complete
- Network-level simulators built and tested
- Script suite with error analysis
- Hardware regime annotations

Action list

- Effect of BB84 repeaters on comparison 
- Vary Charlie's location 
- Practical key rate thresholds (QBER $> 11\%$ cutoff) 
- One-pager: modelling plan & layering 
- Layered parameter analysis 

Modelling Status: Network Modelling

Metropolitan multi-user network model

- N randomly placed metropolitan users; **max connected logical layer**
- K BSM nodes for MDI, all users fibre-connected for BB84
- Passive optical routing for cross-relay MDI pairs (as Q*Bird's)

Simulation experiments

- **Exp 1 — relay sweep:**
 - fix N , vary K
 - relay positions optimised by k -means
 - identify optimal $K = K^*$
 - compare with BB84 for given logical layer
 - per-user pair Monte Carlo sims, as in P2P
- **Exp 2 — user sweep:** fix $K = K^*$, vary N

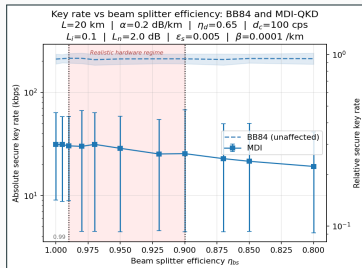
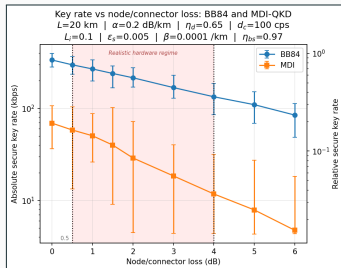
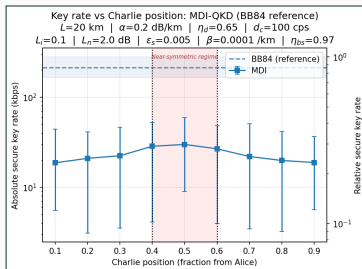
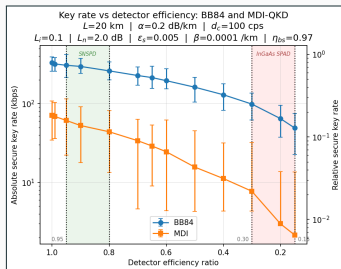
Metrics: average key rate; cost per unit key rate; keygen success rate

Results I: Further P2P Parameters

Swept **10 physical parameters**; realistic hardware regimes identified

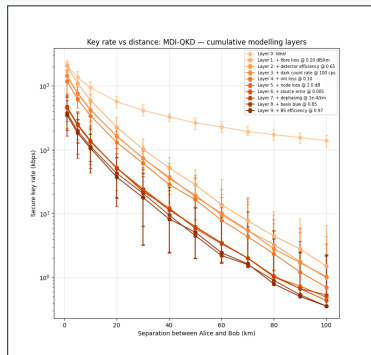
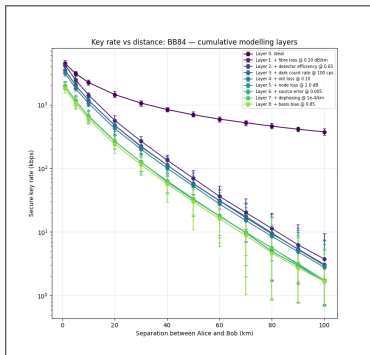
Parameter(s)	Key finding
Fibre length & loss	~Linear key rate drop-off for both protocols
Detector efficiency	MDI - quadratic decline; BB84 - linear
Dark counts, dephasing, source error	Negligible effect in investigated ranges
Node/connector loss & basis bias	Linear declines; both steeper for MDI
Beam splitter efficiency	Shallow MDI decline within realistic range
Charlie placement	Peaks at midpoint; weak symmetric decay

Results I - cont.



Results II: Layered P2P Analysis

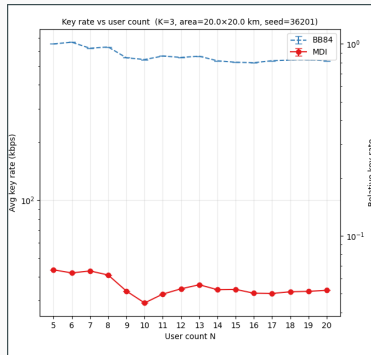
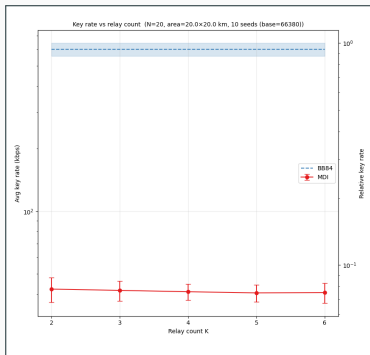
Cumulative layer build-up, ideal $\rightarrow$ full physical model



Node loss applies steepest drop from the prior layer for both. Other parameters cause marginal drops. Ideal conditions depart significantly from realistic.

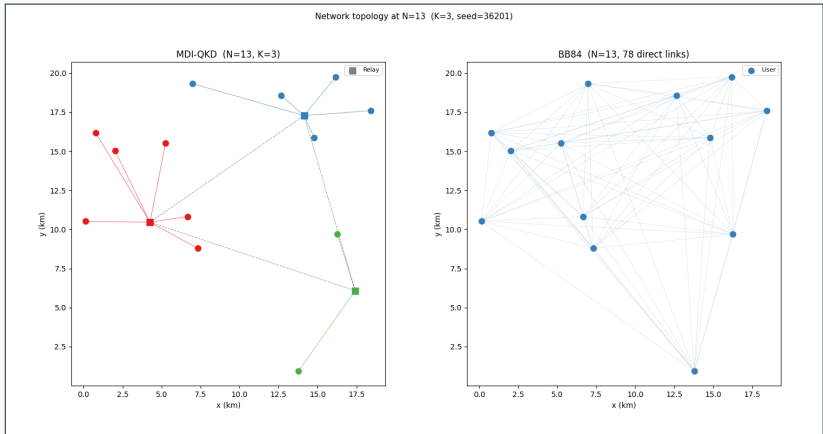
Results III: Early Network-Scale Results

Exp 1 (relay sweep): minimal rate penalty imposed by adding more BSM relays to MDI (example topology next slide). **Exp 2 (user sweep):** both protocols see shallow key rate decline when adding users.



BB84 outperforms MDI in network **more severely** than P2P link.

Results III cont.



Relays placed by k -means clustering for MDI. **Multiple seeds** in single simulation produce distinct networks, rate curves average across these to **eliminate irregularities** in network generation.

Next Steps

Simulation priorities:

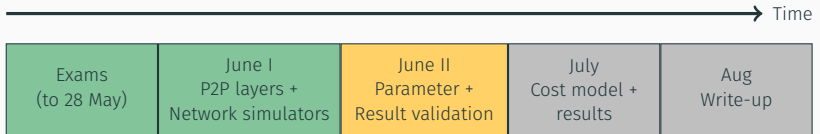
- Finalise hardware parameters from literature
- Overlay published BB84 + MDI curves/data points
- Data persistence for analysis

Analysis priorities:

- Per-component hardware cost markers
- Identify regions where MDI/BB84 performance preferable
- Identify regimes where hardware is preferable

Case Study - microwave spin-ensemble realisation of quantum memories with Prof. John Morton (due 13 Jul 26)

Timeline



- **Remainder of June:** Hardware parameter range selection; literature result overlays; data storage
- **July:** cost model; result collection; Case Study; first draft; Singapore trip
- **August:** Write-up, final figures, submission preparation
- **September:** Viva